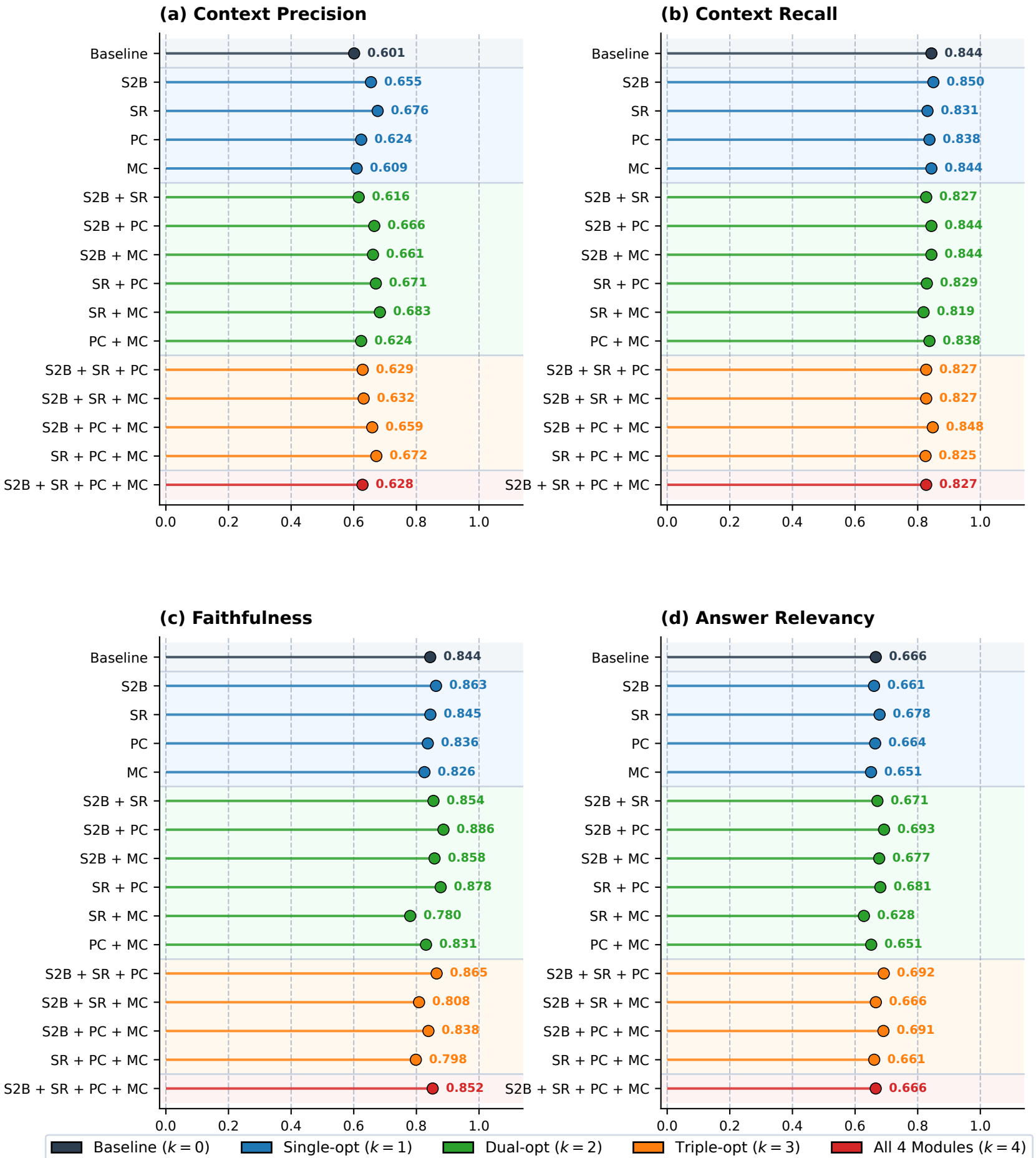
RAGAS Quality Metrics by Pipeline Architecture — Enterprise Closed
(a) Context Precision
Baseline 0.601
S2B 0.655
SR 0.676
PC 0.624
MC 0.609
S2B + SR 0.616
S2B + PC 0.666
S2B + MC 0.661
SR + PC 0.671
SR + MC 0.683
PC + MC 0.624
S2B + SR + PC 0.629
S2B + SR + MC 0.632
S2B + PC + MC 0.659
SR + PC + MC 0.672
S2B + SR + PC + MC 0.628
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.844
S2B 0.850
SR 0.831
PC 0.838
MC 0.844
S2B + SR 0.827
S2B + PC 0.844
S2B + MC 0.844
SR + PC 0.829
SR + MC 0.819
PC + MC 0.838
S2B + SR + PC 0.827
S2B + SR + MC 0.827
S2B + PC + MC 0.848
SR + PC + MC 0.825
S2B + SR + PC + MC 0.827
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.844
S2B 0.863
SR 0.845
PC 0.836
MC 0.826
S2B + SR 0.854
S2B + PC 0.886
S2B + MC 0.858
SR + PC 0.878
SR + MC 0.780
PC + MC 0.831
S2B + SR + PC 0.865
S2B + SR + MC 0.808
S2B + PC + MC 0.838
SR + PC + MC 0.798
S2B + SR + PC + MC 0.852
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.666
S2B 0.661
SR 0.678
PC 0.664
MC 0.651
S2B + SR 0.671
S2B + PC 0.693
S2B + MC 0.677
SR + PC 0.681
SR + MC 0.628
PC + MC 0.651
S2B + SR + PC 0.692
S2B + SR + MC 0.666
S2B + PC + MC 0.691
SR + PC + MC 0.661
S2B + SR + PC + MC 0.666
0.0 0.2 0.4 0.6 0.8 1.0
Baseline (k = 0)
Single-opt (k = 1)
Dual-opt (k = 2)
Triple-opt (k = 3)
All 4 Modules (k = 4)